

Multiple Regression II: Categorical Predictors, Model Comparison, Nonlinearity, and Robust Uncertainty

Jinsong Chen

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Chapter overview

Chapters 7 and 8 fitted lines and additive models and read them as conditional comparisons. Along the way they used tools they could only name: a factor coded as 0/1 columns, a contrast between two categories, a joint test of several added coefficients, an information criterion, a squared predictor, a logged outcome, a covariance estimate that allows unequal spread, and a covariance that respects pupils sitting in schools. This chapter develops each of them once, on the same 98 occupation records, on a small randomized reading study with a pretest, and then on Project STAR, where a randomized design supplies a different warrant. By the end of the chapter you will be able to code and read a factor with several levels, estimate a contrast with the right uncertainty, read an analysis of covariance as a regression on a randomized pretest–posttest study, compare nested models and read AIC and BIC, change a predictor’s form or an outcome’s scale and say what question changed, choose between classical, HC3, cluster-robust, and bootstrap uncertainty for a stated reason, and say what none of these calculations repairs.

A regression report rarely stops at one additive equation. It compares groups that are categories, adjusts a comparison for a baseline score, asks whether a set of added variables matters jointly, ranks candidate models, lets a relationship curve or a scale change, and reports uncertainty calculated in more than one way. Each of these steps has a small mathematics of its own and a temptation of its own: reading a coefficient as “the effect of type”, adding two standard errors, comparing criteria across different outcomes, or treating a robust standard error as a repair for a design problem.

We keep the occupation records of Chapter 8, with their 98 complete rows, income in thousand 1971 Canadian dollars, and the additive model of education, prestige, percentage of women, and type, so that every tool is seen on numbers already read. Two further studies join them where the occupations cannot teach the point: a randomized reading study with a pretest, for the analysis of covariance, and Project STAR, where the uncertainty calculation has to follow the design that produced the data. Keeping the records fixed makes one distinction visible throughout: changing a model, changing a question, and changing an uncertainty calculation are three different acts.

The R code shown is the one call that produces each new quantity; the Computer Lab carries the preparation, the checks and the variations. The type levels remain *bc*, *prof*, and *wc*, the full additive formula remains `income_k ~ education + prestige + women + type`, and every adjusted type comparison names education, prestige, and the percentage of women as its conditioning set. Where Chapter 8’s appendices already worked a tool through on these records (the group means recovered by indicators, the overall *F*, the HC3 covariance, and the Normal likelihood behind AIC and BIC), this chapter points back to that working and adds the new use.

1 Predictors That Are Categories

1.1 One indicator, then several

Chapter 7, Section 3.3, fitted a single 0/1 predictor to 24 invented feedback-by-practice scores. The fitted line passed through the two group means, its slope was the difference between them, 4 with SE 1.6967 and $t = 2.3575$ on 22 df, and that test was the equal-variance two-sample test written as a regression. Nothing about that arrangement is special to two groups.

A factor with J levels enters a regression as $J - 1$ indicator columns. One level is omitted and becomes the **reference**: the intercept is its fitted mean, and each indicator coefficient is a difference from it. With three occupation types and blue-collar as the reference,

$$\hat{y}_g = \hat{\alpha} + \hat{\delta}_{\text{prof}} D_{\text{prof}} + \hat{\delta}_{\text{wc}} D_{\text{wc}}, \quad \hat{\mu}_{\text{bc}} = \hat{\alpha}, \quad \hat{\mu}_{\text{prof}} = \hat{\alpha} + \hat{\delta}_{\text{prof}}, \quad \hat{\mu}_{\text{wc}} = \hat{\alpha} + \hat{\delta}_{\text{wc}}.$$

Two indicators, not three, carry three means. A third column would equal the intercept column minus the other two, so the design matrix would lose full rank and no coefficient would be separately estimable. Which level is omitted is therefore a choice of coordinates, not a choice of model.

The occupation records and models are those of Chapter 8, re-created here so that every later calculation has a stated origin: `m0`, `m1`, and `m2` are the education-only, the education-and-prestige, and the full additive models.

```
prestige <- carData::Prestige
prestige$income_k <- prestige$income / 1000
d <- prestige[!is.na(prestige$type), ]
d$type <- factor(d$type, levels = c("bc", "prof", "wc"))
m0 <- lm(income_k ~ education, data = d)
m1 <- lm(income_k ~ education + prestige, data = d)
m2 <- lm(income_k ~ education + prestige + women + type, data = d)
```

The 98 retained records hold 44 blue-collar, 31 professional, and 23 white-collar occupations, and the full model fits six coefficients, leaving 92 residual degrees of freedom. Complete-case selection is how these rows were chosen; it is not random sampling from a labour market (Fox et al., 2022; Fox & Weisberg, 2019).

1.2 Indicators recover fitted group means

Fit income on type alone and read the coefficients against Figure 1. A blue-collar occupation carries 0 in both indicator columns, a professional one a 1 in the first, and a white-collar one a 1 in the second, so each type's fitted value is the intercept plus that type's coefficient.

```
group_fit <- lm(income_k ~ type, data = d)
coef(group_fit)
```

```
(Intercept)    typeprof    typewc
   5.374136    5.185315   -0.321832
```

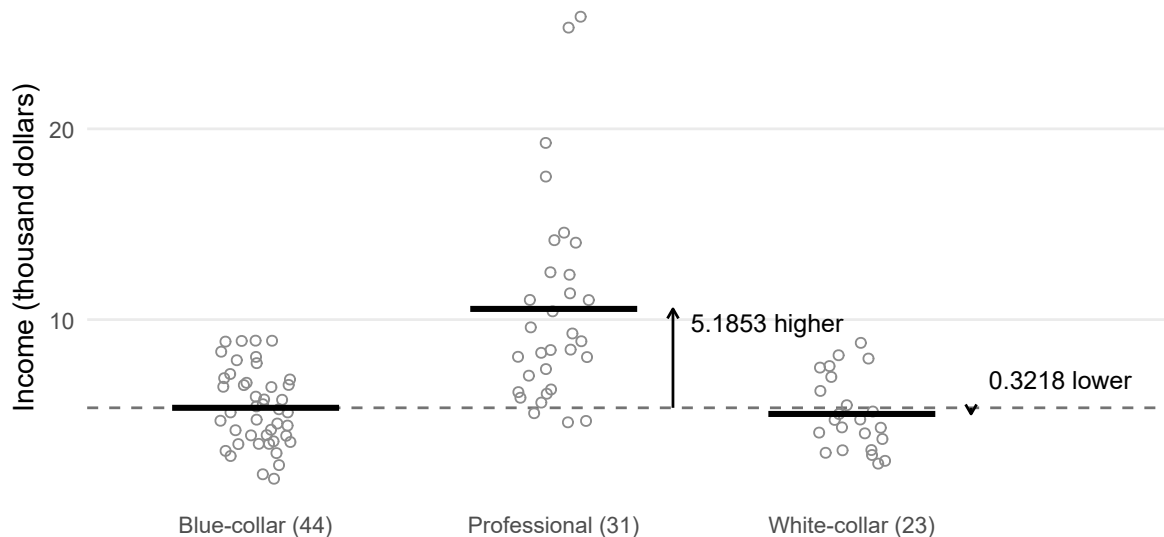


Figure 1: Income of the 98 occupations by type. Bars mark the group means; the dashed line is the blue-collar mean, the intercept, and the arrows are the two indicator coefficients, 5.1853 and -0.3218 .

The coefficients are 5.3741, 5.1853, and -0.3218 . The intercept is the blue-collar mean, the dashed line in the figure, and each coefficient is an arrow from that line to another type's mean: professional occupations average 5.1853 thousand dollars more, white-collar occupations 0.3218 less. The fitted means, 5.3741, 10.5595, and 5.0523, are exactly the observed means, for the reason Chapter 8, Appendix A, gave: with one free parameter per group, least squares returns each group's own mean. A group-only regression is therefore a description of three means with no adjustment in it at all.

The figure also shows what three means leave out. Professional incomes spread from about 4.6 to 25.9 thousand dollars, while the other two types stay below 9; Section 5 returns to that difference in spread.

Keep this description apart from the adjusted comparison of the full model. In `m2` the professional and white-collar coefficients are 0.3242 and 0.2353: they compare fitted incomes at the *same* education, prestige, and percentage of women, not the observed incomes of differently

composed groups. The raw professional advantage of 5.1853 thousand dollars and the adjusted 0.3242 answer different questions, and neither is “the effect of occupational type”.

1.3 Releveling and centering change coordinates, not the fit

Releveling makes a different level the reference. Refit the group-only model with white-collar as the reference:

```
group_fit_wc <- lm(income_k ~ relevel(type, ref = "wc"), data = d)
coef(group_fit_wc)
```

```

              (Intercept)    relevel(type, ref = "wc")bc
              5.052304              0.321832
relevel(type, ref = "wc")prof
              5.507147
```

The coefficients are now 5.0523, 0.3218, and 5.5071: the intercept is the white-collar mean, and the two coefficients measure the blue-collar and professional means from it. The fitted values and residuals are unchanged, as Chapter 8, Appendix A, stated; the same three means are reported in a different coordinate system. The full model behaves the same way. Releveling `m2` to white-collar gives intercept 0.6559, blue-collar -0.2353 , and professional 0.0889, while the education, prestige, and women coefficients stay exactly as they were.

Centering a continuous predictor is the analogous operation: subtracting a constant moves the profile the intercept refers to, and leaves every fitted value, residual, and fixed substantive contrast alone. Two consequences are worth keeping. A coefficient’s size means nothing without its reference; “the professional coefficient” is short for “professional minus whichever level is currently omitted”. And neither releveling nor centering creates information. Chapter 8 documented how thin the joint support across types is at common education and prestige values; changing coordinates adds no observed comparison.

1.4 The one-way analysis of variance of Chapter 4 as a regression

Chapter 4 tested equality of several group means with a one-way analysis of variance. In the group-only regression that null says that both indicator coefficients are zero, so the reduced model is the intercept-only model, whose residual sum of squares is the total sum of squares about the grand mean. Comparing the two models gives

$$F = \frac{(RSS_{\text{reduced}} - RSS_{\text{full}})/2}{RSS_{\text{full}}/(98 - 3)} = \frac{(1734.0792 - 1138.123)/2}{1138.123/95} = 24.8725.$$

The numerator is the reduction in residual sum of squares that the two indicators buy, per indicator; the denominator is the group-only model's residual mean square on its 95 degrees of freedom.

```
anova(lm(income_k ~ 1, data = d), group_fit)
```

Analysis of Variance Table

Model 1: income_k ~ 1

Model 2: income_k ~ type

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	97	1734.1				
2	95	1138.1	2	595.96	24.872	2.057e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

`aov()` returns the same statistic from the same fitted means, the same residual sum of squares, and the same degrees of freedom. The equality is algebraic. It holds whether or not group membership was randomized, so it supplies no causal warrant of its own; Section 7 shows what a randomized design adds. The Chapter 4 cautions travel with the identity. An omnibus F says that the three means are not all equal under the stated reference distribution; it does not say which differ, and any comparison chosen after looking at this result still needs the family calibration of Chapter 4. The next section gives the calculation that a single declared comparison requires.

2 Contrasts and Adjusted Group Comparisons

2.1 A contrast needs coefficient covariance

A **contrast** is a declared linear combination of coefficients, $C = \mathbf{c}^T \boldsymbol{\beta}$, named before the comparison is read. Its estimate is the same combination of the fitted coefficients, and its variance is the corresponding quadratic form in the coefficient covariance matrix $\widehat{\mathbf{V}}$:

$$\widehat{C} = \mathbf{c}^T \widehat{\boldsymbol{\beta}}, \quad \widehat{\text{Var}}(\widehat{C}) = \mathbf{c}^T \widehat{\mathbf{V}} \mathbf{c}, \quad SE(\widehat{C}) = \sqrt{\mathbf{c}^T \widehat{\mathbf{V}} \mathbf{c}}.$$

For a difference of two coefficients the quadratic form reduces to $\text{Var}(b_1) + \text{Var}(b_2) - 2 \text{Cov}(b_1, b_2)$. Take professional minus white-collar in `m2`, whose coefficients are ordered intercept, education, prestige, women, professional, white-collar, so that $\mathbf{c} = (0, 0, 0, 0, 1, -1)^T$.

```

c_vec <- c(0, 0, 0, 0, 1, -1)
V <- vcov(m2)
c(estimate = sum(c_vec * coef(m2)),
  SE = sqrt(drop(c_vec %*% V %*% c_vec)))

```

```

      estimate      SE
0.08886278 1.07999307

```

The estimate is $0.3242 - 0.2353 = 0.0889$ thousand dollars. Its variance assembles the two coefficient variances and their covariance,

$$2.1406 + 0.9698 - 2(0.972) = 1.1664,$$

so the SE is 1.08 and, with $t_{.975,92} = 1.9861$, the 95% interval is $[-2.0561, 2.2338]$.

The covariance matters because the two type coefficients move together. Both are measured from the same blue-collar reference at the same values of the continuous predictors, and their estimated correlation is .6746: a sample that overstates one tends to overstate the other, so their difference varies less than either movement alone suggests. Dropping the covariance would give an SE of 1.7636, and adding the two standard errors, $1.4631 + 0.9848 = 2.4479$, gives a number that is not the standard error of anything. Only the quadratic form is the variance of the declared comparison.

The vector $\mathbf{c}$ is the question written down. One entry set to 1 and the rest to 0 returns an ordinary coefficient with its usual SE. Entries on the type coefficients that sum to zero, with the intercept entry 0, make a contrast in the Chapter 4 sense, and such a contrast names the same comparison whichever level is the reference; that is why Section 1.3 found the same 0.0889 as the professional coefficient of the releveled fit. Declaring the comparison in advance matters as much as computing it correctly: a contrast chosen because it looked large in the fitted coefficients is a selection from a family, and it needs Chapter 4's calibration for that family.

Read the interval for what it is: a working-model benchmark for professional against white-collar occupations at the same education, prestige, and percentage of women. It covers zero, so these records leave the comparison unresolved, and its width does not overcome the thin cross-type support that Chapter 8 documented. An interval can be arithmetically correct and still describe a comparison that the records barely observe.

2.2 Adjusted block tests

A contrast tests one declared combination. A **block test** asks whether a set of coefficients is jointly zero; it is the nested-model comparison of Section 1.4 with a different reduced model. Two blocks are worth testing here, and they are different questions.

```
no_type <- lm(income_k ~ education + prestige + women, data = d)
anova(m1, m2)
anova(no_type, m2)
```

Adding the percentage of women and the two type indicators to `m1` tests three restrictions against the full model's 92 residual degrees of freedom; dropping type from the full model tests two restrictions against the same denominator.

Table 1

Two declared adjusted block comparisons.

Added	df	F	p
Women and type	3	11.5088	< .0001
Type after women	2	0.0319	.9686

The first block is strongly supported: $F(3, 92) = 11.5088$ against a 5% critical value of 2.7036. The second is not: $F(2, 92) = 0.0319$ with $p = .9686$ against a critical value of 3.0954. Neither result says what it is often read as saying. Joint evidence for the first block does not establish that each of the three added columns contributes; the percentage of women could be carrying the whole block. And the second result is not proof that occupational type is irrelevant. It says that, once education, prestige, and the percentage of women are in the equation, the two type indicators add no detectable reduction in residual sum of squares *on these 98 records*, where the types barely overlap on the other predictors.

Two further distinctions matter. `anova(m2)` on a single model gives **sequential** sums of squares in formula order: each line tests a term given only the terms above it, so the answers depend on the order in which the formula was written, and only the last line coincides with the corresponding block test. And Welch-type and robust-covariance tests are different procedures with different reference distributions; Section 5 gives the robust version of this type block.

2.3 The analysis of covariance as regression

A randomized study with a baseline measurement asks the group question in a sharper form. Baumann and colleagues randomly assigned 68 fourth-grade pupils in one school to three reading-instruction conditions (Baumann et al., 1992): DRA, a directed reading activity

from the basal programme; DRTA, a directed reading-thinking activity; and think-aloud strategy instruction. Before and after the lessons, each pupil took an error-detection test of comprehension monitoring. The `carData::Baumann` distribution holds the 66 pupils with a recorded first posttest, 22 per group. The analysis here uses the first pretest and the first posttest, labels the groups DRA, DRTA, and Think-aloud, and takes DRA as the reference; the original investigators analysed further outcomes and both pretests.

The **analysis of covariance** adds the pretest to the group comparison as one more predictor with a single slope:

$$Y_{ij} = \mu_j + \beta(X_{ij} - \bar{X}) + \varepsilon_{ij}, \quad \bar{Y}_j^{\text{adj}} = \bar{Y}_j - \hat{\beta}(\bar{X}_j - \bar{X}).$$

Here Y_{ij} and X_{ij} are the posttest and pretest of pupil i in group j . The group terms μ_j shift the line up or down, and the pretest enters once, so every group's line has the same slope β . An **adjusted mean** is a group's fitted mean at a stated pretest value. At the grand pretest mean $\bar{X}$ it is the group's observed posttest mean $\bar{Y}_j$ moved along the common slope by the group's pretest departure from $\bar{X}$, which is the second equation.

```
reading <- carData::Baumann
reading$group <- factor(reading$group,
  levels = c("Basal", "DRTA", "Strat"),
  labels = c("DRA", "DRTA", "Think-aloud"))
ancova <- lm(post.test.1 ~ group + pretest.1, data = reading)
coef(ancova)
```

(Intercept)	groupDRTA	groupThink-aloud	pretest.1
-0.5966478	3.6265538	2.0361644	0.6931872

```
predict(ancova, interval = "confidence",
  newdata = data.frame(group = levels(reading$group),
    pretest.1 = mean(reading$pretest.1)))
```

	fit	lwr	upr
1	6.188185	5.143481	7.232889
2	9.814739	8.779995	10.849482
3	8.224349	7.181274	9.267425

Read Figure 2 from each filled symbol, which marks a group's observed pretest and posttest means, to the open symbol on the vertical line, which marks its adjusted mean. The DRA group happened to start highest, 0.7121 points above the grand pretest mean of 9.7879. Following its line to the grand mean lowers its posttest mean by 0.6932×0.7121 , from 6.6818 to 6.1882.

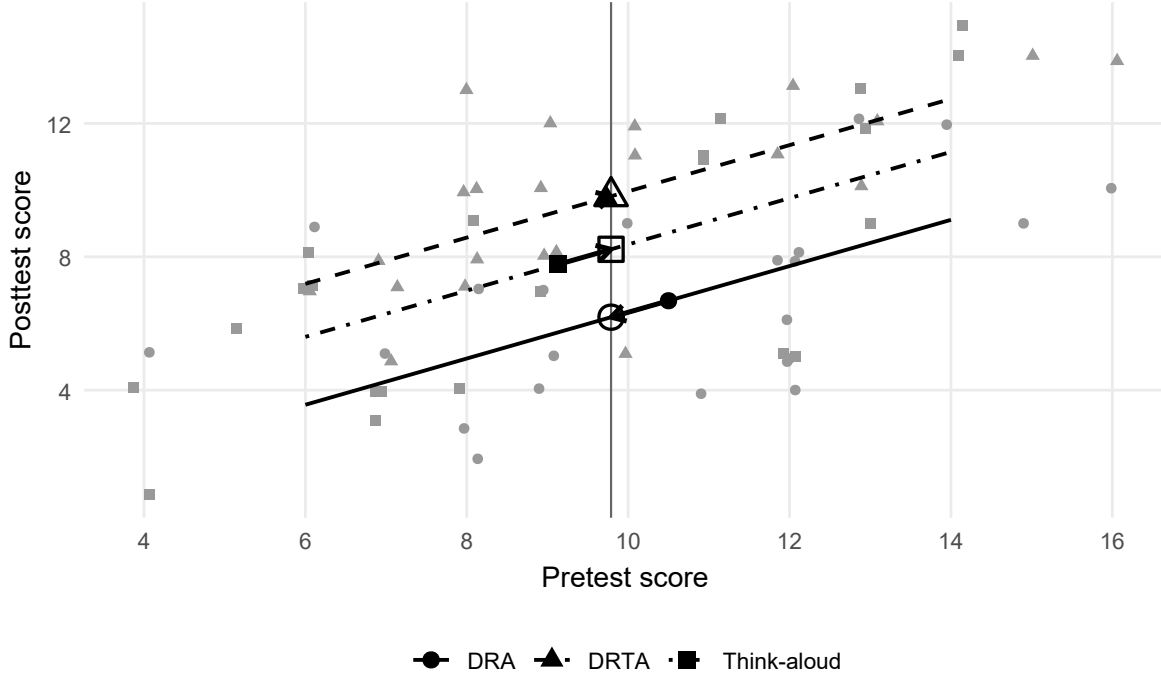


Figure 2: Posttest against pretest for the 66 Baumann pupils. The parallel lines are the common-slope fit over the common pretest support 6–14; each filled symbol marks a group’s observed pretest and posttest means, and its arrow follows the group’s line to the grand pretest mean 9.7879, where the open symbol marks the adjusted mean.

Think-aloud started 0.6515 points below the grand mean and rises from 7.7727 to 8.2243, while DRTA, almost exactly at the grand mean, barely moves, from 9.7727 to 9.8147. The fit gives the common slope 0.6932 (SE 0.1015), the differences from DRA of 3.6266 (SE 0.7362) for DRTA and 2.0362 (SE 0.7450) for Think-aloud, and a residual mean square of 5.894 on 62 degrees of freedom. Tested together after the pretest, the two group coefficients give $F(2, 62) = 12.1689$, $p < .0001$, and the printed adjusted means carry the 95% intervals [5.1435, 7.2329], [8.7800, 10.8495], and [7.1813, 9.2674].

Three conditions carry that reading. The adjusted means need a **common reference**, here the grand pretest mean. That reference must lie inside every group’s observed pretest range, the **common support**: the groups’ pretests run from 4 to 16, 6 to 16, and 4 to 14, so the lines are drawn over 6 to 14, and an adjusted comparison read outside that band would rest on extrapolation. And the comparison needs a stated **averaging population**: for this additive linear model, predicting every group at the grand pretest mean gives the same values as predicting every group over the pupils’ common pretest distribution and averaging. Because the lines are parallel, the adjusted differences, 3.6266 and 2.0362, are the same at any pretest value inside the common support; the adjusted means themselves are not, which is why the

reference value is always stated.

2.4 Posttest only, gain score, and covariance adjustment as one regression

Three analyses are common for a randomized pretest–posttest study: compare the posttest means, compare the gains (posttest minus pretest), or adjust for the pretest as above. They are one regression that differs only in the pretest coefficient γ :

$$Y_i - \gamma X_i = \alpha + \delta_{\text{DRTA}} D_{\text{DRTA},i} + \delta_{\text{TA}} D_{\text{TA},i} + \varepsilon_i, \quad \gamma = 0, 1, \text{ or estimated.}$$

With $\gamma = 0$ the model compares posttest means, with $\gamma = 1$ it compares gains, and with γ estimated it is the analysis of covariance. In R, `offset()` enters the pretest with its coefficient fixed at 1:

```
post_only <- lm(post.test.1 ~ group, data = reading)
gain <- lm(post.test.1 ~ group + offset(pretest.1), data = reading)
```

Table 2

Three analyses of the Baumann records as one regression.

Analysis	Pretest coefficient	DRTA minus DRA (SE)	Think-aloud minus DRA (SE)	Residual SD (df)
Posttest only	0	3.0909 (0.9614)	1.0909 (0.9614)	3.1885 (63)
Gain score	1	3.8636 (0.7779)	2.4545 (0.7779)	2.5799 (63)
Analysis of covariance	0.6932, estimated	3.6266 (0.7362)	2.0362 (0.7450)	2.4278 (62)

Under random assignment the expected pretest difference between groups is zero, so all three analyses estimate the same average difference in posttest scores. Strictly, random assignment covers the 68 pupils assigned, while these analyses use the 66 with a recorded first posttest; the argument carries over to the 66 if, for example, the two posttests went missing for reasons unrelated to what those pupils would have scored, a complete-record condition that the design does not guarantee and the records cannot check. The three analyses differ in precision and in how this sample’s chance pretest imbalance enters. Here the pretest means are 10.5000, 9.7273, and 9.1364, so DRA started with a head start of 0.7727 points over DRTA, about a quarter of the pretest SD of 3.0205 across all 66 pupils. The size of that gap is what matters. A significance test of it would add nothing: random assignment already makes any baseline gap a chance gap, so such a test would examine a null hypothesis known to be true.

The gap accounts exactly for the spread of the estimates in Table 2. Each DRTA estimate is the posttest-only difference plus γ times the DRA group's pretest head start, $3.0909 + \gamma(10.5 - 9.7273)$, which gives 3.0909, 3.8636, and 3.6266 for the three values of γ (3.6266 from full precision; 3.6265 from the rounded inputs).

The last column shows why the estimated coefficient is usually preferred. The pretest predicts the posttest with a slope near 0.7, so fixing its coefficient at 0 leaves the pretest's predictive information in the error (residual SD 3.1885), and fixing it at 1 over-corrects (2.5799). Estimating it leaves the smallest residual SD, 2.4278, and the smallest standard errors. The choice among the three belongs in the analysis plan, made before the outcomes are seen; picking whichever estimate looks best after fitting all three turns three defensible analyses into one selected result.

2.5 What adjustment cannot supply

A common slope is a condition, not a finding. The check adds the group-by-pretest product block, which lets each group have its own slope, and compares the two named models on the same records:

```
varying_slopes <- lm(post.test.1 ~ group * pretest.1, data = reading)
anova(ancova, varying_slopes)
```

Analysis of Variance Table

Model 1: `post.test.1 ~ group + pretest.1`

Model 2: `post.test.1 ~ group * pretest.1`

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	62	365.43				
2	60	347.97	2	17.461	1.5054	0.2302

The product block gives $F(2, 60) = 1.5054$, $p = .2302$, with group slopes of 0.4663 (DRA), 0.6868 (DRTA), and 0.8768 (Think-aloud). These 66 records do not distinguish the two models, which is not the same as showing that the slopes are equal. When slopes do differ, an adjusted difference depends on the pretest value at which it is read; Chapter 10 develops that varying-slope model as an interaction.

What adjustment supplies depends on how the groups were formed. In groups formed by random assignment, the pretest only sharpens a comparison that the design already warrants. In observational groups, a fitted comparison at a common covariate value cannot balance what was not measured; a covariate measured with error under-adjusts, leaving part of the baseline difference in place (Chapter 14, Section 5.1); and a covariate recorded after the groups formed can remove part of the effect being estimated.

Lord’s paradox makes the point with intact groups that differ at baseline (Lord, 1967): the gain-score analysis asks whether the groups changed by different amounts, the covariance analysis asks whether pupils with the same pretest ended differently, and the two can reach opposite conclusions from the same records, partly through the regression toward the mean of Chapter 7, Section 7. Neither analysis says which question is the causal one; the design does, and Chapter 14 returns to what a design supplies that adjustment cannot.

3 Compare Nested Models

3.1 The extra-sum-of-squares statistic

Sections 1.4 and 2.2 used one statistic. Let the reduced model be the full model with q coefficients set to zero, both fitted to the same records and the same outcome. Then

$$F = \frac{(RSS_R - RSS_F)/q}{RSS_F/(n - p)}.$$

The numerator is the extra sum of squares that the added block buys, per coefficient it spends. The denominator is the fuller model’s residual mean square, the same s^2 that appears in every classical standard error. Under the Normal linear model with independent common-variance errors, the ratio has an F reference with q and $n - p$ degrees of freedom.

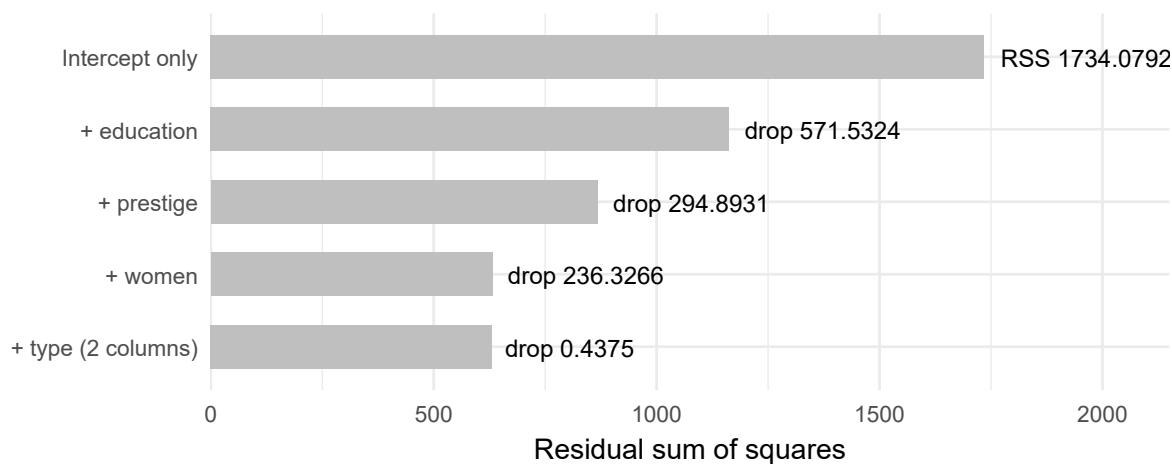


Figure 3: Residual sum of squares as columns are added to the occupation model on the same 98 records; each label gives the drop that the added column or block buys.

Figure 3 shows that bargain for the occupation models. Education buys 571.5324 of the total 1734.0792, and prestige a further 294.8931. The percentage of women then buys 236.3266

for one coefficient, while the two type columns buy only 0.4375 for two. Set against the full model’s residual mean square of 6.8575, the type block’s 0.4375 spread over two coefficients gives $F(2, 92) = 0.0319$, and the women-and-type block’s 236.764 over three gives 11.5088, the entries of Table 1. Section 1.4’s test was the same comparison with the intercept-only model as the reduced model.

The subtraction means something only when the two models use the same rows and the same outcome, and when the reduced model is obtainable from the fuller one by restrictions; Chapter 8’s complete-case selection is why every comparison in this chapter runs on the same 98 records. The **overall** regression test is the special case whose reduced model is the intercept alone, so that all $p - 1$ non-intercept coefficients are set to zero. Chapter 8, Appendix A, gave its formula and read it from `summary()`: the full model gives $F(5, 92) = 32.1747$, and the education-only model on the same 98 records gives $F(1, 96) = 47.1956$. Placed beside the general statistic, the two values are easy to misread. The larger one is not evidence that `m0` is the better model, because the two statistics test different restrictions in different models, each against the intercept alone. Neither verifies the mean function, the error model, or the independence of the rows, and neither says how a fitted rule would perform on records it has not seen, which is a separate question with a separate calculation (Chapter 13).

3.2 Restrictions in matrix form

The same statistic has a compact matrix form, worth one pass because it makes clear what “the same records” is doing in every comparison above. Let $\mathbf{X}$ hold n rows and p coefficient columns including the intercept, with full column rank. Minimizing $(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$ gives the normal equations $\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}$ and hence $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$, $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ with $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$, and $\hat{\mathbf{V}} = s^2(\mathbf{X}^\top\mathbf{X})^{-1}$. For q independent linear restrictions $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$,

$$F = \frac{(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})^\top [\mathbf{R}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{R}^\top]^{-1} (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})}{q s^2} \sim F_{q, n-p}.$$

For `m2`, $\mathbf{X}$ has six columns and rank 6: the two type indicators do not reproduce the intercept column, so every coefficient is separately estimable. Choosing the two rows of $\mathbf{R}$ that select the type coefficients, with $\mathbf{r} = \mathbf{0}$, returns 0.0319, the adjusted type block test of Section 2.2, without fitting a second model.

Two practical points follow. `lm()` does not form $(\mathbf{X}^\top\mathbf{X})^{-1}$; it uses a QR decomposition, which is numerically better behaved, so the displayed equation explains what the estimator *is*, not how R computes it (Fox, 2016; Fox & Weisberg, 2019). And because $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ is a projection onto the column space of $\mathbf{X}$, any full-rank recoding of a factor spans the same space and leaves the fitted vector alone, which is Section 1.3’s releveling result stated geometrically. Appendix A carries the derivation.

3.3 A local Normal likelihood and information criteria

Nested F tests compare two models when one contains the other. Information criteria rank models that need not be nested, using a different quantity: the maximized likelihood, penalized for the number of fitted parameters.

Chapter 8, Appendix B, built that likelihood for these occupation models by carrying Chapter 6's binomial likelihood, for 32 successes in 50 trials, over to outcomes treated as independent Normal draws around the regression. Three of its results carry the reading here. Maximizing the Normal likelihood over the coefficients reproduces least squares. Its scale estimate divides the residual sum of squares by n , $\sqrt{RSS/n} = 2.5373$, where the residual SD reported by `lm()` divides by the residual degrees of freedom, $\sqrt{RSS/(n-p)} = 2.6187$. And AIC adds $2k$ to $-2\log \hat{L}$ while BIC adds $k \log n$, where k counts the fitted coefficients **and** the residual variance, so `m2` has $k = 7$, not 6. One call returns the maximized log-likelihood and both criteria for a fitted model:

```
c(logLik = logLik(m2), AIC = AIC(m2), BIC = BIC(m2))
```

logLik	AIC	BIC
-230.3020	474.6040	492.6988

For `m2` it returns -230.3020 , 474.6040 , and 492.6988 . Chapter 8, Appendix B, set the same criteria for the three nested models side by side: AIC 526.5052, 499.8328, and 474.6040 for `m0`, `m1`, and `m2`, and BIC 534.2601, 510.1727, and 492.6988. Smaller is preferred under either criterion, and both prefer `m2` here.

What this chapter adds is how to use them as tools for comparing models. BIC charges the same seven parameters $7 \log 98 = 32.0948$ instead of AIC's 14, because at $n = 98$ the factor $\log n = 4.585$ exceeds 2. The two criteria can therefore disagree, and a disagreement is information about how strongly the extra parameters are being charged, not a tie to be broken by preference. The conditions for reading them are strict and easy to breach. The candidates must be fitted to the same records, with the same outcome on the same scale, under the same likelihood convention; Section 4.2 shows what happens when the scale changes. Above all, neither criterion is a held-out error. Both are calculated on the records that fitted the model, and both are comparative rather than absolute: a smaller AIC says one candidate is preferred to another, not that either describes the records well (Akaike, 1974; Schwarz, 1978). Chapter 11 develops likelihood fully, for models where least squares is no longer available.

4 Change the Form of a Predictor or the Scale of the Outcome

4.1 A polynomial term

Chapter 8, Section 4.3, refitted the full model with a squared prestige term as a disclosed sensitivity check on functional form. Here that refit is the worked example. A squared term is an ordinary extra column, `I(prestige^2)`, and the model is still linear in its coefficients. What changes is the reading:

$$E(Y | x, \dots) = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots, \quad \frac{\partial E(Y)}{\partial x} = \beta_1 + 2\beta_2 x, \quad x^* = -\frac{\beta_1}{2\beta_2}.$$

The predictor no longer has a single slope. It has a slope that depends on its own value, $\beta_1 + 2\beta_2 x$, and a turning point x^* where that slope is zero.

```
m2q <- lm(income_k ~ education + prestige + I(prestige^2) + women + type,
  data = d)
anova(m2, m2q)
```

Analysis of Variance Table

Model 1: income_k ~ education + prestige + women + type

Model 2: income_k ~ education + prestige + I(prestige^2) + women + type

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	92	630.89				
2	91	587.41	1	43.481	6.736	0.01101 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The quadratic coefficient is 0.0024 with SE 0.0009 and $t = 2.5954$; the one-restriction block test gives $F(1, 91) = 6.736$ with $p = .011$; and the criteria move from 474.6040 and 492.6988 to 469.6058 and 490.2855. Adding the term also changes the education coefficient from 0.1164 to -0.0298 , a reminder that coefficients in an additive model are defined relative to the other columns present.

Figure 4, the component-plus-residual display of Chapter 8 with the fitted curve added, shows what the squared term claims. With linear coefficient -0.0754 and quadratic coefficient 0.0024, the curve's slope is 0.0697 thousand dollars per prestige point at prestige 30, 0.1665 at 50, and 0.2633 at 70: income rises with prestige throughout the records, and rises faster at the top, where the straight line of the additive model falls below several high-prestige occupations. The turning point, 15.5906, lies below the smallest observed prestige of 17.3, so the descending branch of the parabola is entirely outside the records. Reporting a “negative prestige coefficient”

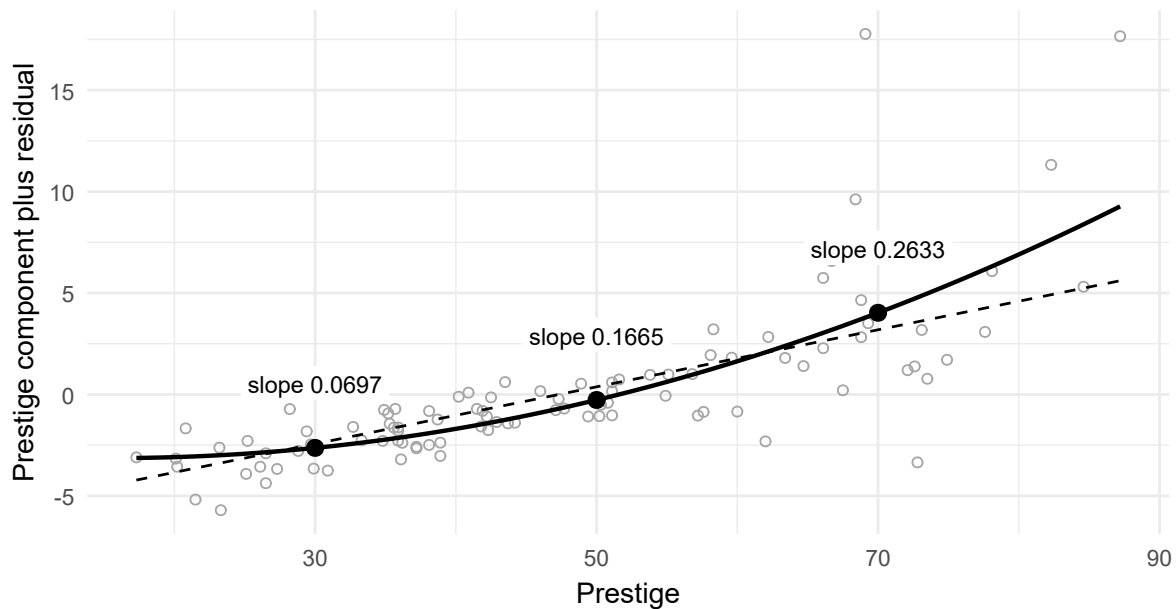


Figure 4: Component-plus-residual display for prestige in the model with a squared prestige term. The solid curve is the fitted quadratic component and the dashed line the linear component of the additive model; the labels give the curve's slope at prestige 30, 50 and 70.

from this model would misread the coordinate -0.0754 , which is the slope at prestige zero, a value no occupation has.

Centering repairs that coordinate problem without changing the curve, because it is a change of coordinates. Prestige and its square correlate .9849; centering prestige at its mean, 47.3276, before squaring lowers that correlation to .3216. The quadratic coefficient and its SE stay 0.0024 and 0.0009, and every fitted value stays the same. The linear coefficient becomes 0.1536 with SE 0.0347: it now estimates the slope at the mean prestige, a different quantity from -0.0754 , the slope at prestige zero, whose SE is about 2.6 times as large. The smaller SE belongs to a slope read in the middle of the records, not to a sharper estimate of the old one. Keep the status of the exercise straight: this is a disclosed sensitivity check on functional form, declared before the refit, and its result is that a modest curvature is detectable on these records. It is not a search across candidate terms for the smallest p -value, and a p -value read after such a search would not mean what it appears to mean.

4.2 A log-transformed outcome

Changing a predictor's form keeps the question. Changing the outcome's scale changes the question, and that difference is the point of this subsection.

$$\log Y_i = \beta_0 + \sum_j \beta_j x_{ij} + \varepsilon_i, \quad 100(e^{\beta_j} - 1) \text{ percent per unit of } x_j.$$

A coefficient in a log-income model is an additive difference in expected log income. Exponentiating turns it into a multiplicative difference in geometric mean income under that model, and $100(e^{\beta} - 1)$ expresses that multiplier as a percentage. The geometric mean is not the arithmetic mean, so this is not automatically a statement about average income in dollars.

```
m_log <- lm(log(income) ~ education + prestige + women + type, data = d)
100 * (exp(coef(m_log)[c("education", "prestige", "women")]) - 1)
```

```
education    prestige      women
0.7660700    2.0964846 -0.8429719
```

Table 3

Log-model contrasts on a geometric-mean percent scale.

Term	Log coefficient	Percent	Lower	Upper
education	0.0076	0.7661	−4.2041	5.9941
prestige	0.0207	2.0965	1.4351	2.7622
women	−0.0085	−0.8430	−1.0189	−0.6667

One more prestige point is associated with about 2.0965 percent higher geometric-mean income at the same education, percentage of women, and type, with interval [1.4351, 2.7622] percent; one more percentage point of women with about −0.843 percent. The two type coefficients, −0.0137 and 0.0708 on the log scale, are −1.3584 and 7.3341 percent. The intervals in Table 3 are the classical log-scale intervals with both endpoints transformed, which is legitimate because the transformation is monotone; they inherit the log-scale working model and supply no new warrant.

The criteria will not travel across this change. This model's AIC is 8.091 against `m2`'s 474.604, and the difference says nothing at all: the two likelihoods are densities of different response variables, so their values are not on a common scale. The discipline to keep is simple: fix the outcome and the question inside any comparison, and treat a scale change as a new question rather than as data cleaning (Fox & Weisberg, 2019).

4.3 Coefficient scales: standardized coefficients and unique contributions

Two summaries are often asked for once several predictors are in an equation, and both are more limited than their names suggest:

$$\beta_j^* = \hat{\beta}_j \frac{s_{x_j}}{s_y}, \quad \Delta R_j^2 = R_{\text{full}}^2 - R_{\text{without } j}^2.$$

A **standardized coefficient**, or beta weight, rescales a partial slope into outcome-SD units per predictor SD. A **drop-one** R^2 reports how much of the explained variation disappears when one predictor is removed from the full model. The beta weights are 0.0757, 0.5685, and -0.3971 for education, prestige, and the percentage of women, and education's drop-one R^2 is .0007: removing it from the full model costs essentially nothing in fit, the same message the education coefficient's wide interval carried in Chapter 8.

Both quantities are rescalings of an in-sample fit, and neither measures importance. A standardized coefficient depends on the predictor's observed spread in these particular records, so a restricted range shrinks it for reasons that have nothing to do with the relationship. It also differs from the standardization over named profiles that Chapter 8 used, which averages fitted values over a declared distribution rather than dividing by an SD. The drop-one change describes only the *unique* contribution: because predictor information overlaps, the separate drop-one values need not sum to the joint R^2 , and a predictor that duplicates another can show a near-zero value while mattering. Neither number establishes a causal ranking or predicts held-out performance.

5 Robust Uncertainty Under Unequal Spread

5.1 The HC3 covariance

Every classical standard error so far assumed one common error variance for all 98 occupations. Figure 5 shows why that assumption is doubtful here.

The residuals of professional occupations spread about four times as widely as the others: their SD is 4.3242 thousand dollars against 1.0437 for blue-collar and 1.0243 for white-collar occupations. A single s^2 averages these very different spreads, so it overstates the uncertainty contributed by some rows and understates it for others.

The remedy is already in hand. Chapter 7, Section 6.1, first used a robust standard error on the simple regression line, where the education slope's SE moved from 0.1270 to 0.1491. Chapter 8 then applied the same HC3 estimator to the full model: its Section 3.3 computed the education interval with `sandwich::vcovHC()`, and its Appendix B displayed the estimator, which keeps the design component outside and replaces the single s^2 inside by each row's own

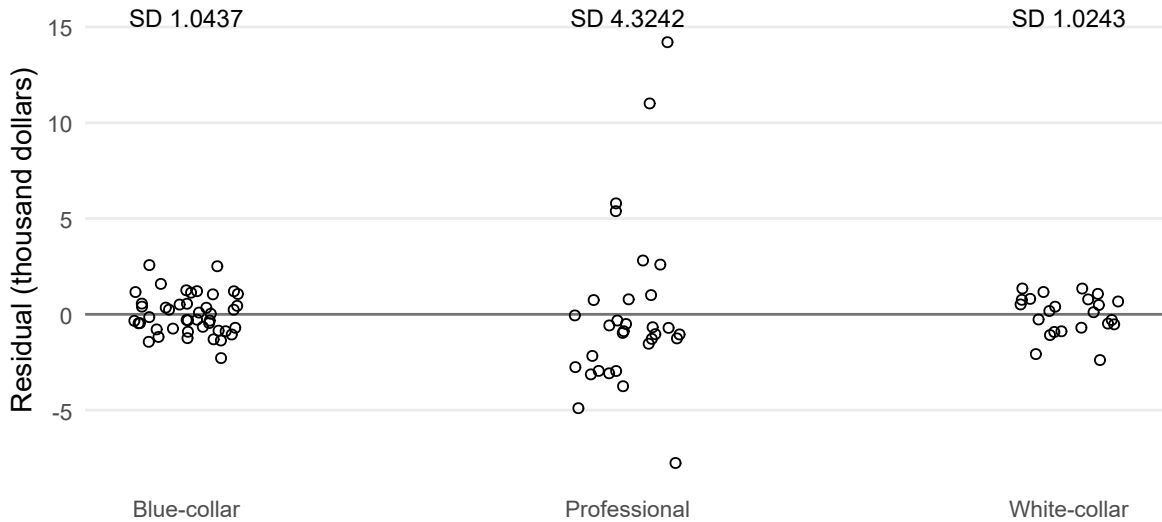


Figure 5: Residuals of the full occupation model by type, with each type's residual SD.

leverage-adjusted squared residual, and set classical and HC3 standard errors side by side for all six coefficients (Long & Ervin, 2000; MacKinnon & White, 1985; White, 1980; Zeileis, 2004). Appendix A of this chapter shows where the estimator comes from.

What the figure adds is a reason for the pattern in that table. The professional coefficient's SE grows by about 40%, from 1.4631 to 2.0928, as the wide professional spread predicts, while the white-collar SE grows by only about 7%, from 0.9848 to 1.0548. The education SE grows from 0.2752 to 0.3823, and its 95% interval on the robust covariance is $[-0.6429, 0.8758]$: wider than the classical one, with the same reading of an imprecisely estimated comparison. Every coefficient, fitted value, and residual stays exactly what it was, because HC3 replaces a covariance estimate, not a fit.

The percentage of women is the instructive case: its SE *falls* from 0.0097 to 0.0079. A robust covariance is not a penalty applied to every coefficient. It is a different estimate of the same target, and where the classical common-variance assumption happened to overstate a coefficient's variability, the robust version is smaller. Reporting only the coefficients whose intervals widened, or switching estimators after seeing which choice gives the preferred conclusion, turns a stated method into a selection.

5.2 A block test with a robust covariance

The restriction statistic of Section 3.2 needs only a coefficient vector and a covariance matrix. Substituting the HC3 covariance $\widehat{\mathbf{V}}_{\text{HC3}}$ for $\widehat{\mathbf{V}}$ gives an approximate Wald version of the same block test. For the type block it gives 0.0345, with $p = .9661$, against the classical 0.0319

and .9686: here the substitution changes nothing of substance. The reason is that the type coefficients sit close to the restriction, each within a quarter of its classical SE of zero, so both statistics are near zero; the SEs would have to shrink about tenfold before either statistic reached the .05 critical value, 3.0954. A change of covariance matters most when a block statistic lies near its critical value.

The reference distribution is where the two differ in kind. The classical statistic has an exact $F_{2,92}$ distribution under the stated Normal linear model. The robust version has only an approximate reference, justified as the sample grows, and F with the same degrees of freedom is a convention rather than a derivation. Quoting the robust p -value to several digits, or treating it as the same object as the classical one because both are labelled F , overstates what the calculation delivers.

5.3 What HC3 does not repair

HC3 has one job. Under a maintained linear conditional mean and independent observations, it estimates the coefficient covariance without assuming a common error variance.

It does not repair curvature: if the mean function is wrong, the coefficients estimate a different quantity (Appendix A names it), and a robust covariance describes that quantity's variability more honestly without making it the quantity of interest. It does not repair dependence: rows that are not independent break the estimator's own derivation, and Section 7 shows a case where a robust row-level SE and a naive one agree closely while both understate the uncertainty by nearly half. It does not repair selection or missing outcomes, which change what the records represent, and it does not supply causal identification, which comes from design or from an argument about the conditioning set. Error in measuring a predictor also pulls its slope toward zero, by roughly the predictor's reliability when it is the only predictor, and no covariance estimator repairs it (Cohen et al., 2003); independent mean-zero error in the outcome, by contrast, widens the residual spread and the standard errors but leaves the slope's target unchanged. Chapter 14, Section 5.1, develops attenuation. Nor does HC3 create joint support: the type contrast's robust SE of 1.1827 is still the uncertainty of a comparison the records barely observe.

The practical response is to say which estimator was used and why, and to keep the answer's limits attached to it. Chapter 8's advice holds here: report the named comparison, its form and influence sensitivities, and the covariance calculation, rather than treating any one of them as a pass/fail certificate.

6 Resample the Records

6.1 Why resample, and which scheme

Every uncertainty statement so far came from a formula: a coefficient covariance matrix, classical or robust, propagated through a linear combination. Resampling offers a different approximation. Refit the model to many resamples of the records, and read the spread of the refitted estimates directly (Davison & Hinkley, 1997; Efron, 1979; Efron & Tibshirani, 1993). Where the formula and the resampling distribution disagree, the disagreement is itself informative.

The choice of what to resample is an assumption, not a default.

- **Case resampling** draws whole records with replacement and refits. It inherits whatever unequal spread the rows contain, but it requires the resampling unit to match the sampling unit, with units independent and exchangeable.
- **Residual resampling** holds the design fixed, resamples rescaled residuals, and adds them back to the fitted values. It assumes identically distributed errors, the assumption HC3 exists to relax.
- **The wild bootstrap** also holds the design fixed but multiplies each rescaled residual by an independent mean-zero, unit-variance draw, so every record keeps its own error scale.

This chapter uses case resampling, for two stated reasons. The occupation predictors were not designed: no one assigned education or prestige to an occupation, so a joint-sampling view of the rows is the natural one. And the residual spread differs by type, as Figure 5 showed, so the identically distributed errors behind residual resampling are not available (Freedman, 1981). Appendix B compares the other two schemes on the same targets.

6.2 Case resampling for the full model

Resample the 98 rows with replacement, refit `m2`, and keep the education slope. The seed and the number of resamples B belong in the record, so both appear in the code; B counts bootstrap resamples here, as it counted simulation repetitions in Chapter 5.

```

set.seed(90909)
B <- 2000
boot_education <- replicate(B, {
  rows <- sample.int(nrow(d), replace = TRUE)
  coef(lm(income_k ~ education + prestige + women + type,
    data = d[rows, ]))["education"]
})
c(SE = sd(boot_education), quantile(boot_education, c(.025, .975)))

```

```

      SE      2.5%      97.5%
0.3698895 -0.6829568  0.7439809

```

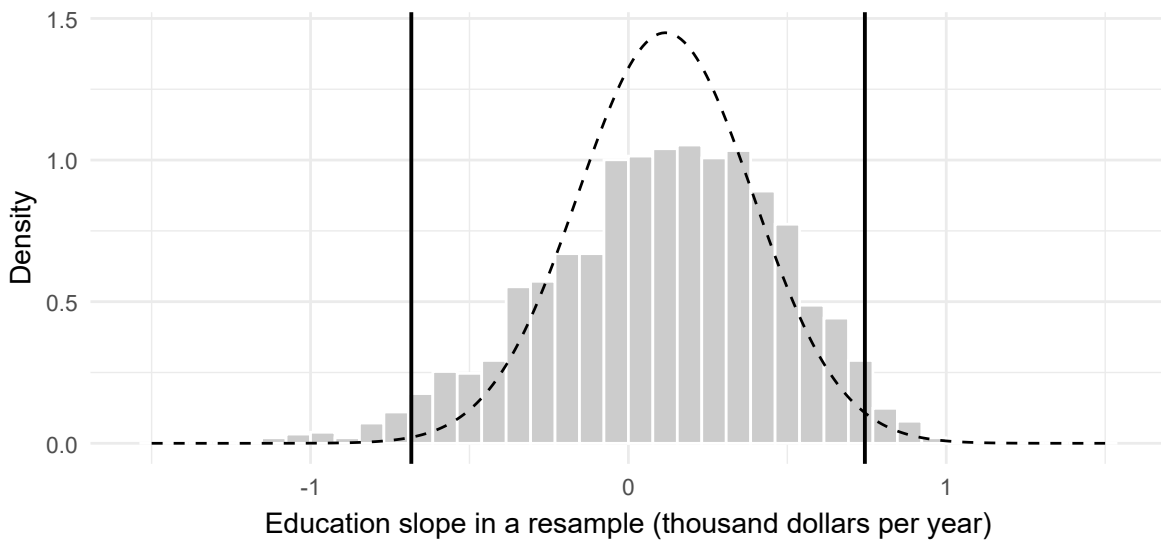


Figure 6: The 2,000 case-resampled education slopes (seed 90909). Solid lines mark the percentile interval; the dashed curve is the Normal distribution with the classical SE 0.2752.

Figure 6 is the resampling distribution itself. The 2,000 refitted slopes spread more widely than the Normal curve built from the classical SE: their SD, the bootstrap SE, is 0.3699, and their middle 95%, the percentile interval, runs from -0.683 to 0.744 . The same resamples give the other two targets of Table 4, the professional-minus-white-collar contrast and the professional coefficient. A resample could omit an entire occupational type, and its refit would then lose a type coefficient; here all 2,000 resamples contained all three types, so all 2,000 refits are usable, and that count is part of the result rather than an assumption.

Table 4

Model-based, HC3, and case-bootstrap uncertainty for three targets in the full occupation model (seed 90909, $B = 2000$).

Target	Estimate	Classical SE	HC3 SE	Bootstrap SE	Percentile 2.5%	Percentile 97.5%
Education slope	0.1164	0.2752	0.3823	0.3699	−0.6830	0.7440
prof — wc	0.0889	1.0800	1.1827	1.1187	−1.9613	2.4313
prof coefficient	0.3242	1.4631	2.0928	1.9926	−3.1416	4.5443

Read the table one row at a time. The education slope’s bootstrap SE of 0.3699 sits between the classical 0.2752 and the HC3 0.3823: three calculations, one conclusion that this comparison is imprecise. The type contrast’s bootstrap SE of 1.1187 exceeds the classical 1.0800, with percentile interval $[-1.9613, 2.4313]$. The professional coefficient shows the largest movement, from a classical 1.4631 to 1.9926, close to the HC3 2.0928; with only 31 professional occupations spreading as widely as Figure 5 shows, a resample’s composition matters, and the formula that assumes a common error variance is the most optimistic of the three.

Agreement between HC3 and the bootstrap is not independent confirmation. Both are sensitivity procedures applied to the same 98 records, and both inherit the same limitations of those records; agreement means the two approximations behave similarly here, not that the answer has been replicated. Monte Carlo variation also means that the last decimal of a bootstrap summary carries no weight: a different seed would move it, which is exactly why the seed and $B = 2000$ are reported.

6.3 Which interval, and what resampling does not repair

Two intervals come directly from the resampling distribution. The **percentile interval** takes its 2.5th and 97.5th percentiles, as in Figure 6. The **normal-approximation interval** takes the original estimate plus and minus 1.96 bootstrap standard errors. The two agree when the resampling distribution is roughly symmetric; here the three distributions have skewness -0.3891 , 0.2635 , and 0.3785 , read by their sign as in Chapter 2, Section 2.1: the education slopes have the slightly longer left tail visible in the figure, the other two a slightly longer right tail. The asymmetry is mild enough that the two kinds of endpoint agree closely. For a strongly skewed or biased resampling distribution, **bootstrap- t** and **BCa** intervals are the higher-order alternatives; Appendix B defines both in words.

What resampling cannot do matters more than which interval to choose. Drawing occupations with replacement treats them as independent and exchangeable draws from one population, and there are stated reasons to doubt both. The occupations share one historical labour market, where common conditions can move related occupations’ incomes together; the 98 rows were chosen by complete records rather than sampled at random from a population of occupations;

and their measurements are aggregates over workers. Resampling does not repair the thin cross-type support, correct the measurement problems Chapter 8 documented, or convert an observational comparison into a causal one. And if a design contained clusters, strata, or weights, the resampling unit would have to follow the design rather than the rows, which is the subject of the next section.

7 Dependence in the Design: Project STAR

Every calculation so far has used observational records or one small experiment, and every qualification has been about what those records cannot support. This section changes the design. Project STAR randomized kindergarten pupils to class types within their schools, which supplies a warrant no amount of adjustment can manufacture, and a dependence structure that the uncertainty calculation must follow.

7.1 A different design supplies a different warrant

The question is specific:

Among eligible kindergarten pupils in the participating Tennessee schools, what was the mean reading-score difference under assignment to a small rather than a regular class?

The outcome is `readk`, an end-of-kindergarten reading scale score. The field `stark` records the kindergarten class-type condition. Reading its small-versus-regular comparison as an assignment comparison rests on the documented experimental protocol, not on the column name or on the `lm()` call: random assignment occurred **within** participating schools (Finn & Achilles, 1990; Mosteller, 1995). The regular-with-aide arm is outside this declared two-arm comparison and is excluded by design, not dropped for convenience.

Three limits belong with the question from the start. Class **type** is not a measured headcount, so this analysis estimates no dose response in pupils per room. The distributed field does not independently verify each pupil's original randomized assignment; an assignment-effect reading additionally requires linking the recorded type to the assignment record and checking switches and implementation deviations, which this teaching reanalysis does not supply. And the scientific target is the pupil-weighted mean assignment effect for the eligible cohort in participating schools, while a common coefficient from a complete-record model is an estimator that needs further links, a common effect across schools or an explicit weighting argument, to reach that target.

The data are distributed as `AER::STAR`, accompanying *Applied Econometrics with R* and following the version used by Stock and Watson (Kleiber & Zeileis, 2008; Stock & Watson,

2007). They are loaded from the package rather than redistributed here, and the analysis is retrospective rather than preregistered.

7.2 Separate eligibility from missingness

Two different reasons for a record’s absence from an analysis are easy to merge and must be kept apart. Some distributed rows belong to pupils who entered the study after kindergarten: they have no kindergarten condition at all, and their absence is **structural ineligibility**. Other rows belong to eligible kindergarten pupils whose reading score or covariates were not recorded: that is **missingness**, and it can distort a comparison. Only the variables a calculation actually uses may enter its complete-record rule; in particular, a missing mathematics score (`mathk`) must not exclude an otherwise usable reading record.

Table 5
STAR eligibility and reading-analysis completeness.

Stage	<i>n</i>
Distributed records	11,598
Kindergarten condition recorded	6,325
Eligible small or regular	4,094
Eligible, reading observed	3,745
Complete primary records	3,735
Primary regular	2,001
Primary small	1,734
Primary schools	79

Of 11,598 distributed records, 5,273 have no kindergarten condition; they are later entrants, not 5,273 failed measurements. The remaining 6,325 have a recorded kindergarten arm, and across all three arms 536 lack a reading score. Restricting to the two declared arms leaves 4,094 eligible records, of which 349 lack reading, 188 in the regular arm and 161 in the small arm, leaving 3,745 with an observed reading score, 2,006 regular and 1,739 small.

Similar missing percentages in the two arms are reassuring but prove nothing: the absent scores could differ systematically within either arm, and no observed value can reveal that. Complete-record inference here requires an assumption that, given the declared covariates and within arm and school, observation does not depend on the reading score that would have been recorded. Stating the assumption is not verifying it. Among the observed-reading records across all arms, 38 lack at least one of lunch status, gender, ethnicity, and teacher experience; within the two focal arms, ten do. The primary common-covariate sample therefore has 3,735 pupils, 2,001 regular and 1,734 small, in 79 schools, and the observed-reading sample keeps all 3,745 focal records as a declared sensitivity, not as an alternative headline.

7.3 The binary slope recovers the observed arm means

Section 1.1's identity applies unchanged. With $D_i = 1$ for the small arm and 0 for regular,

$$\widehat{Y}_i = \hat{\alpha} + \hat{\delta}D_i, \quad \hat{\alpha} = \bar{Y}_{D=0}, \quad \hat{\delta} = \bar{Y}_{D=1} - \bar{Y}_{D=0} = 440.5474 - 434.7323 = 5.8151.$$

On the 3,745 observed-reading records the regression slope reproduces the arm difference of 5.8151 reading points exactly, as Chapter 7's binary predictor did on 24 pupils. The identity is algebraic and requires no randomization; what randomization adds is the reason to read the difference as an assignment comparison.

Two cautions attach to the number itself. Under independent common-variance rows, the slope's t test equals the pooled two-sample test, here $t = 5.6039$; that row-level reference ignores the fact that pupils sit in schools, which Section 7.5 addresses. And raw arm means weight each school by its own arm-specific pupil count, so the pooled difference equals the pupil-weighted cohort target only under common assignment probabilities and comparable observation patterns, or after explicit standardization to target school weights.

7.4 Compare two specifications on identical primary rows

The adjusted model adds free-lunch status, gender, ethnicity, teacher experience, and school indicators. School indicators respect the assignment blocks; the remaining covariates are the declared adjustment specification, not a substitute for randomization.

```
star_unadjusted <- lm(readk ~ stark, data = star_df)
star_primary <- lm(readk ~ stark + lunchk + gender + ethnicity +
  experiencek + schoolidk, data = star_df)
```

On the common 3,735 primary rows, the unadjusted difference is 5.8955 and the adjusted coefficient is 6.6575. Adjustment can improve precision when the added variables are prognostic for reading, and school indicators match the blocks in which assignment happened. It is not needed to remove baseline confounding that a successfully implemented randomization already prevents, and it does not automatically preserve a causal target: timing, selection, and the fitted comparison still need scrutiny. This is Section 2.5's point about randomized groups, on a larger design.

The earlier 5.8151 comes from 3,745 rows and must not serve as the same-row unadjusted baseline for 6.6575. Comparing 5.8151 with 6.6575 would confound two changes, a change of records and a change of specification, and the whole discipline of this chapter is to keep those apart. The same-row pair to compare is 5.8955 against 6.6575. Similar unadjusted and adjusted coefficients are consistent with successful randomization; they are not proof that implementation, missingness, or weighting problems are absent.

7.5 Cluster uncertainty follows the design

Pupils are the outcome units. Schools are the randomization blocks and the approximately independent units for the uncertainty calculation; the schools themselves were not assigned to arms. Chapter 3, Section 2.3, showed that the uncertainty of a school-blocked comparison follows its within-school comparisons, and Figure 7 shows those comparisons for STAR.

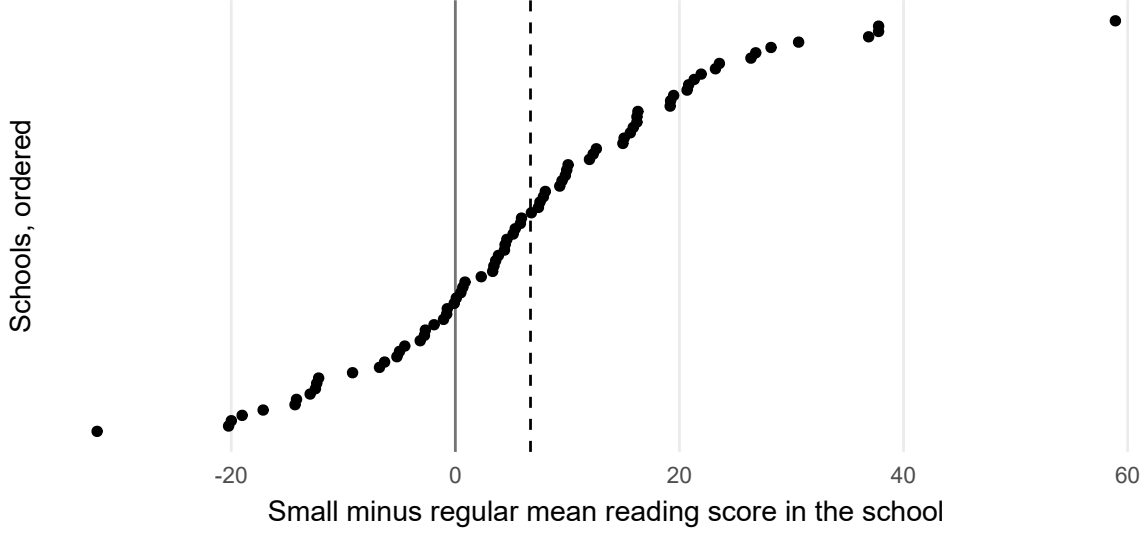


Figure 7: Small-minus-regular mean reading difference in each of the 78 schools that observed both class types, ordered by size. The dashed line is their unweighted mean, 6.7094.

In the 78 schools that observed pupils in both arms, the small-class advantage ranges from about -32 to 59 reading points, and it is negative in 25 schools. Pupils in one school share whatever made their school's difference large or small: its teachers, its intake, its classrooms. The 3,745 pupils therefore do not supply 3,745 independent pieces of information about the arm difference; the schools do. A rough check makes the scale visible. An average of 78 independent differences with SD 15.4192 has SE $15.4192/\sqrt{78} = 1.7459$, the size of the clustered standard errors below and far from the row-level 1.0377.

For school g , form its score $\mathbf{s}_g = \mathbf{X}_g^\top \mathbf{e}_g$ by summing the residual contributions **within** the school before taking an outer product. The CR1/HC1 cluster-robust covariance is then

$$\widehat{\mathbf{V}}_{\text{CR1}} = c_G (\mathbf{X}^\top \mathbf{X})^{-1} \left(\sum_{g=1}^G \mathbf{s}_g \mathbf{s}_g^\top \right) (\mathbf{X}^\top \mathbf{X})^{-1}, \quad \mathbf{s}_g = \mathbf{X}_g^\top \mathbf{e}_g, \quad c_G = \frac{G}{G-1} \frac{n-1}{n-p}.$$

The within-school cross-products are exactly what distinguishes this from treating every pupil as independent, and c_G is the finite-cluster correction. With $G = 79$ schools, intervals use a t_{78} critical value, 1.9908 (Zeileis et al., 2020).

```
star_V <- sandwich::vcovCL(star_primary, cluster = star_df$schoolidk,
  type = "HC1", cadjust = TRUE)
sqrt(star_V["starksmall", "starksmall"])
```

```
[1] 1.697409
```

Table 6

Class-type comparisons and school-clustered intervals.

Model	<i>n</i>	Estimate	SE	Interval
Observed: unadjusted	3,745	5.8151	1.8476	[2.1368, 9.4935]
Primary: unadjusted	3,735	5.8955	1.8491	[2.2143, 9.5767]
Primary: adjusted	3,735	6.6575	1.6974	[3.2782, 10.0368]
Observed: school-adjusted	3,745	6.6273	1.7797	[3.0842, 10.1703]

Table 6 holds four distinct calculations, each with its own records and its own specification; each row's interval uses the school identifiers from that model's own rows. Changing the model or its records is a different act from changing the covariance calculation, and the table keeps them separable.

Set the clustered SE beside Section 5's robust one. For the same 3,745-record unadjusted comparison, treating pupils as independent rows gives SE 1.0377; HC3 gives 1.0416, barely different; school clustering gives 1.8476, nearly twice as large. Unequal spread across pupils was not this design's problem. Dependence within schools was, and no row-level robust covariance addresses it. The adjusted primary model shows the same pattern: its row-level SE is 0.9161 and its HC3 SE 0.9509, against the clustered 1.6974.

These are cluster-asymptotic approximations with 79 clusters, not exact randomization inference and not a multilevel model. A residual plot cannot choose the independence unit; the design does. And a count of 79 clusters does not certify the approximation: the balance of information across clusters and the influence of particular schools also matter.

7.6 Read the bounded answer, then evaluate its support

A chapter-authored report can now state:

On 3,735 complete two-arm kindergarten records, the common adjusted small-versus-regular class-type coefficient was 6.6575 reading points, with approximate school-clustered interval [3.2782, 10.0368]. Its causal mean-effect interpretation relies on documented within-school randomization, a verified link from recorded

class type to original assignment, suitable complete-record selection, independent-school uncertainty, and a common-effect or target-matching weighting argument. It concerns assigned class type, one kindergarten reading outcome, and participating Tennessee schools, not realized headcount or a portable policy dose response.

Read the report's functions before judging it, applying Chapter 1's argument components as Chapters 7 and 8 did. The sample audit and the model code carry **STAT-E** content; the design and covariance reasoning supplies **STAT-J**; the estimate and its clustered interval supply **STAT-I**; and the first sentence is the bounded **STAT-C** answer. **SUB-E** situates the pupils, the test, the cohort, and the school context; **SUB-J** adds the assignment record together with the selection and weighting qualifications; **SUB-I** reads the coefficient as assignment-related only under that support; and **SUB-C** bounds the substantive claim. These are linked functions within one argument, not two arguments.

Then separate what is *stated* from what is *demonstrated*. The report states the selection and weighting conditions; stating them does not establish that they hold. The randomization protocol is documented in the study's own publications, not shown by this coefficient table. The link from the recorded class-type field to the original assignment is asserted here and not verified. Where a necessary link cannot be assessed, the honest move is to report the limitation rather than to invent it or to quietly substitute a different target.

The design literature documents implementation and interpretation issues that a single coefficient cannot settle (Chetty et al., 2011; Hanushek, 1999; Krueger, 1999; Nye et al., 2000). Nor does this result transport by itself to nonparticipating schools, another period, another outcome, or another education system. Those limits belong inside the answer, not in a list appended after an unrestricted claim, and a crossed significance boundary is not what makes a result (Gelman et al., 2020; Gelman & Stern, 2006). Chapter 14 returns to dependence as a design fact, reuses this calculation as its example, and asks what a design supplies that adjustment cannot.

8 Reading Bridge

A factor enters a regression as indicators, and its fitted means, contrasts, and omnibus test are the analysis of variance written as a model; a contrast's uncertainty needs the covariance between coefficients, not the sum of two standard errors. The analysis of covariance is the same regression with a pretest added: the posttest-only, gain-score, and adjusted analyses differ only in the pretest coefficient, and adjustment sharpens a randomized comparison without supplying what randomization supplies. Nested models compare on the same records and the same outcome by an extra-sum-of-squares F , and information criteria rank them by penalized in-sample fit, never by held-out error. A squared predictor gives a comparison that depends on the predictor's value; a logged outcome changes the question to a geometric-mean scale, and its criteria no longer speak to the original one. HC3, the case bootstrap, and a clustered covariance each change the uncertainty calculation for a stated reason and leave the fitted coefficients alone; none repairs curvature, missing outcomes, weighting, or the absence of a design. Project STAR showed the uncertainty following the design: schools, not pupils, were the approximately independent units, and the difference was nearly a factor of two.

Chapter 10 asks a different question of the same equations. Every comparison in this chapter was the same at every value of the other predictors, which is what "additive" means, and Section 2.5's common-slope check was the first test of it. Chapter 10 lets a comparison itself vary with another predictor, and develops how to read, probe, and report such an interaction.

Key terms

Use this short reference to revisit the concepts introduced here. The Part One Concept Reference gives the broader reference and the principal home of each concept; indicator coding, partial slopes, coefficient covariance, and HC3 are explained in Chapters 7 and 8, and regression toward the mean in Chapter 7.

Table 7

Selected concepts for the regression toolbox.

Term and common alternatives	Meaning
ANCOVA (analysis of covariance)	A regression comparing groups while including a quantitative covariate with one common slope; its adjusted group means are read at a stated covariate value within common support. The posttest-only, gain-score and ANCOVA analyses are one regression whose pretest coefficient is 0, 1 or estimated; adjustment cannot supply what random assignment supplies.

Term and common alternatives	Meaning
Homogeneity of regression slopes (parallel slopes)	ANCOVA's condition that the outcome-covariate slope is the same in every group, checked by testing the group-by-covariate product block; when the slopes differ, the adjusted difference depends on the covariate value at which it is read.
Nested-model F test (extra sum of squares; block test)	Compares a reduced model with a fuller one fitted to the same records and outcome: the RSS reduction per added coefficient, divided by the fuller model's residual mean square, referred to F with the number of restrictions and the fuller model's residual df. It tests the added block jointly, not each coefficient separately.
Information criteria (AIC; BIC)	Penalized in-sample fit: $-2 \log \widehat{L}$ plus $2k$ (AIC) or $k \log n$ (BIC), with k counting the coefficients and the residual variance. Smaller is preferred among models fitted to the same records and outcome scale; neither is held-out error.
Polynomial term (quadratic term; curvature)	A power of a predictor added as a further column so that the fitted mean can curve. The predictor's comparison then depends on its value; centering before squaring reduces collinearity without changing the fitted curve.
Log-transformed outcome (geometric-mean scale)	Fitting the logarithm of the outcome, so that a coefficient is an additive difference in expected log outcome and $100(e^\beta - 1)$ percent is a multiplicative difference in geometric means. A changed outcome scale is a changed question, and its criteria are not comparable with the original scale's.
Standardized coefficient (beta weight)	A partial slope rescaled by the predictor and outcome SDs, in outcome-SD units per predictor SD, with the other predictors held fixed. It differs from standardization over named profiles and does not measure causal importance.
Bootstrap (case resampling; percentile interval)	Refitting the model to many resamples of the records drawn with replacement and using the spread of the refitted estimates as an SE, or their middle 95% as a percentile interval. It treats the resampled units as independent and exchangeable; it does not repair dependence, thin support, measurement error, or an observational design.
Cluster-robust standard error (school-clustered covariance)	A coefficient covariance that sums residual contributions within clusters, such as schools, before combining them, allowing dependence within clusters under an approximation that treats clusters as independent. It changes uncertainty, not the coefficient, and does not repair missing outcomes or weighting.

Further reading

- The OpenIntro multiple-predictor and model-selection tutorial gives concrete indicator interpretations and criterion-based reading. We use its equation-first device without adopting automatic stepwise selection.
- The ISLR regression lab and the ISLR resampling lab connect fitting, factors, criteria, and resampling in one workflow (James et al., 2021).
- Fox (Fox, 2016) and Kutner and colleagues (Kutner et al., 2005) develop the linear-model algebra, the restriction statistic, and the diagnostics behind this chapter’s matrix section.
- Efron and Tibshirani (Efron & Tibshirani, 1993) and Davison and Hinkley (Davison & Hinkley, 1997) are the standard bootstrap references, including the bootstrap- t and BCa intervals named in Section 6.3.
- The sandwich documentation specifies the HC and CR covariance calls used here (Zeileis et al., 2020).

Exercises

Use the supplied numbers; rounding within 0.02 is acceptable unless a different precision is requested. These prompts do not require the appendices.

Exercise 1: Read indicator coefficients (Foundational)

A group-only regression of income (thousand dollars) on occupation type has observed group means 5.3741 (blue-collar, 44 records), 10.5595 (professional, 31), and 5.0523 (white-collar, 23). (1) With blue-collar as reference, give the intercept and the two indicator coefficients. (2) Give the intercept and both coefficients with white-collar as reference. (3) Explain why the fitted values are the same under both codings. (4) The full additive model’s professional coefficient is 0.3242. Explain why it differs from the professional coefficient in (1) and why neither is “the effect of occupational type”.

Exercise 2: A contrast needs covariance (Foundational)

In the full additive model the professional and white-collar coefficients are 0.3242 and 0.2353, with estimated variances 2.1406 and 0.9698 and covariance 0.9720, on 92 residual df. (1) The professional-minus-white-collar contrast is 0.0889. Calculate its variance, and explain why the covariance enters with a minus sign and what a positive covariance does to the result. (2) The contrast’s SE is 1.08 and its 95% interval is $[-2.0561, 2.2338]$. A student instead adds the two coefficient SEs (1.4631 and 0.9848). State what that number is and why it is wrong. (3) Explain what the interval does and does not say about the two types.

Exercise 3: A nested comparison from residual sums of squares (Foundational)

On the same 98 records, the model with education and prestige has RSS 867.6537 on 95 df, and the full model adding the percentage of women and the two type indicators has RSS 630.8897 on 92 df. The 5% critical value of $F(3, 92)$ is 2.7036. (1) Calculate the F statistic for the added block. (2) State the decision at 5% and what it does and does not say about the three added columns. (3) Explain why the sequential sums of squares from a single `anova()` call on the full model would not answer this question.

Exercise 4: What information criteria compare (Foundational)

The full additive model has maximized log-likelihood -230.302 on 98 records, with AIC 474.6040 and BIC 492.6988; the model with education and prestige only has AIC 499.8328 and BIC 510.1727. (1) The criteria count seven fitted parameters for the full model. Say what the seven are, and why BIC charges the full model more for them than AIC does ($\log 98 = 4.585$). (2) Which model do the criteria favour, and on what condition is that comparison meaningful? (3) A log-income version of the full model has AIC 8.091. Explain why this number cannot be compared with 474.604.

Exercise 5: Percent and curvature (Integrative)

Two changed models of the occupation records raise different questions. (1) In a log-income model the prestige coefficient is 0.0207. Give the percent difference in geometric-mean income per prestige point and say what “geometric mean” adds to the reading. (2) In an income model with a squared prestige term, the fitted slope is 0.0697 thousand dollars per prestige point at prestige 30, 0.1665 at 50 and 0.2633 at 70, and the turning point is at prestige 15.5906; prestige ranges from 17.3 to 87.2. Explain why the model has no single prestige slope, and what the position of the turning point tells you. (3) State in one sentence what the curve claims about income and prestige over the observed range, and one reason that claim is a sensitivity check rather than a discovery.

Exercise 6: Choose the uncertainty calculation (Integrative)

In Project STAR the small-minus-regular kindergarten reading difference on 3,745 observed records is 5.8151 points. Three standard errors are available: 1.0377 treating pupils as independent, 1.0416 with HC3, and 1.8476 clustering pupils within their 79 schools. Treating pupils as independent gives the 95% interval $[3.7807, 7.8496]$. (1) Using $t_{.975, 78} = 1.9908$, give the interval under the clustered calculation and compare its width with the first. (2) Say which interval a report should use and why, naming the unit at which assignment happened, and explain why HC3 barely changes the first standard error. (3) Name two problems that none of the three calculations repairs.

Exercise 7: One randomized study, three analyses (Integrative)

In the Baumann records, 66 pupils in three randomized instruction groups have a pretest and a posttest. With DRA as reference, the DRTA difference is 3.0909 (SE 0.9614) from the posttest alone, 3.8636 (SE 0.7779) from the gain score, and 3.6266 (SE 0.7362) from the analysis of covariance, whose pretest coefficient is 0.6932; the DRA and DRTA pretest means are 10.5000 and 9.7273. (1) Write the three analyses as one regression and give the pretest coefficient each uses. (2) Explain why all three estimate the same average difference under random assignment, and show how the pretest gap accounts for the differences between the three estimates. (3) Say which analysis you would report, and when that choice has to be made. (4) Suppose the groups had been intact classes that differed at the pretest. Explain what changes, naming the paradox that can arise.

Appendix A: Matrix Geometry, Restrictions, and Sandwich Constructions

This appendix derives the results that Sections 3.2, 5.2, and 7.5 used, including the HC3 estimator that Chapter 8, Appendix B, stated, and names the target least squares estimates when a linear mean is only an approximation. It is reference material: the exercises do not require it, and the main route is complete without it. The Computer Lab carries the numerical reproductions of each construction.

Projection and the normal equations

Let $\mathbf{X}$ be $n \times p$ with full column rank, so $\mathbf{X}^\top \mathbf{X}$ is invertible. The least-squares criterion is $S(\boldsymbol{\beta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$. Expanding gives $S = \mathbf{y}^\top \mathbf{y} - 2\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{y} + \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta}$, and differentiating with respect to $\boldsymbol{\beta}$ yields the normal equations

$$\mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{y}, \quad \text{equivalently} \quad \mathbf{X}^\top (\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}}) = \mathbf{0}.$$

The second form is the geometry: the residual vector is orthogonal to every column of $\mathbf{X}$. Hence $\hat{\mathbf{y}} = \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{H} \mathbf{y}$ with $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$, and $\mathbf{H}$ is symmetric and idempotent, the orthogonal projection onto the column space of $\mathbf{X}$. Its trace is p , which is why the leverages h_{ii} sum to the number of fitted coefficients; for `m2` they sum to 6.

Two consequences used earlier follow immediately. Any full-rank recoding $\mathbf{X}\mathbf{A}$, with $\mathbf{A}$ invertible, spans the same column space, so $\mathbf{H}$ and $\hat{\mathbf{y}}$ are unchanged: releveling a factor or centering a predictor cannot alter a single fitted value. Adding a genuinely new column enlarges the space and can change every fitted value. And under the classical error model, $\hat{\mathbf{V}} = s^2(\mathbf{X}^\top \mathbf{X})^{-1}$ with

$s^2 = RSS/(n - p)$, which is where every classical standard error in this chapter came from. Solving the normal equations directly for `m2` reproduces `lm()`'s coefficients to about 10^{-13} .

General linear restrictions

Let $\mathbf{R}$ be $q \times p$ of rank q and consider $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$. Since $\widehat{\boldsymbol{\beta}}$ has covariance $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ under the classical model, the vector $\mathbf{R}\widehat{\boldsymbol{\beta}} - \mathbf{r}$ has covariance $\sigma^2 \mathbf{R}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{R}^\top$. Standardizing by that covariance and dividing by q gives the statistic of Section 3.2, which has an exact $F_{q, n-p}$ reference under Normal errors, because its numerator and denominator quadratic forms are independent chi-squares.

It is the same number the extra-sum-of-squares formula produces. Minimizing $S(\boldsymbol{\beta})$ subject to $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$ with a Lagrange multiplier gives the restricted estimator $\widetilde{\boldsymbol{\beta}} = \widehat{\boldsymbol{\beta}} - (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{R}^\top [\mathbf{R}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{R}^\top]^{-1} (\mathbf{R}\widehat{\boldsymbol{\beta}} - \mathbf{r})$, a correction applied only in the directions the restriction touches. Substituting it into the criterion shows that the increase in residual sum of squares is

$$RSS_R - RSS_F = (\mathbf{R}\widehat{\boldsymbol{\beta}} - \mathbf{r})^\top [\mathbf{R}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{R}^\top]^{-1} (\mathbf{R}\widehat{\boldsymbol{\beta}} - \mathbf{r}),$$

which is exactly the numerator of the restriction statistic. Fitting the reduced model and differencing, or evaluating the quadratic form in the unrestricted fit, are two routes to one number; for the two type coefficients of `m2` both return 0.0319. Replacing $s^2(\mathbf{X}^\top \mathbf{X})^{-1}$ by a robust covariance in the middle of the expression gives the Wald statistic of Section 5.2. The algebra survives the substitution; the exact reference distribution does not.

The HC3 sandwich

Chapter 8, Appendix B, stated the HC3 estimator; this subsection shows where it comes from. Write the coefficient error as $\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon}$. Under independent rows with possibly different variances σ_i^2 , its exact covariance is

$$(\mathbf{X}^\top \mathbf{X})^{-1} \left(\sum_i \sigma_i^2 \mathbf{x}_i \mathbf{x}_i^\top \right) (\mathbf{X}^\top \mathbf{X})^{-1},$$

a *sandwich* with the design component as bread and the error-scale component as filling. The classical estimator substitutes a common s^2 , which collapses the filling to $s^2 \mathbf{X}^\top \mathbf{X}$. The HC family substitutes each row's own squared residual instead. Because $E(e_i^2) = \sigma_i^2(1 - h_{ii})$ under the classical model, the raw residual understates the error at high-leverage rows; HC3 corrects by dividing by $(1 - h_{ii})^2$, which is deliberately slightly conservative and performs well in small samples (Long & Ervin, 2000; MacKinnon & White, 1985). Built directly from these pieces, the estimator reproduces `sandwich::vcovHC(m2, type = "HC3")` to about 10^{-12} .

The CR1 cluster sandwich

Cluster robustness changes the filling, not the bread. Partition the rows into G clusters and let $\mathbf{X}_g$ and $\mathbf{e}_g$ hold cluster g 's rows and residuals. The cluster score $\mathbf{s}_g = \mathbf{X}_g^\top \mathbf{e}_g$ sums the within-cluster contributions *before* the outer product is taken, so the estimator of Section 7.5 retains the within-cluster residual correlations that the independent-row filling discards. If every cluster held one row, the filling would reduce to the HC0 form, which is why a row-level robust SE is the special case that assumes away the dependence.

The asymptotics are in G , not in n : 3,745 pupils in 79 schools supply 79 approximately independent pieces of information for this covariance, which is why the clustered SE of 1.8476 is so much larger than the row-level 1.0377 and why a t_{G-1} reference is used. For the observed-reading model the finite-cluster factor is $c_G = 1.0131$, and building the estimator from the school scores reproduces `sandwich::vcovCL(..., type = "HC1", cadjust = TRUE)` to about 10^{-13} .

What least squares estimates when the mean is not linear

A linear regression is often fitted where the conditional mean of the outcome is not exactly linear in the predictors, as Section 4.1's curved prestige component shows. Least squares then still estimates a defined target: the **best linear projection**, the coefficients that minimize expected squared error over the population the records represent,

$$(\beta_0^*, \beta_1^*) = \arg \min_{a,b} E\{(Y - a - bX)^2\}, \quad \beta_1^* = \frac{\text{Cov}(X, Y)}{\text{Var}(X)},$$

$$\beta^* = \{E(\mathbf{x}\mathbf{x}^\top)\}^{-1} E(\mathbf{x}Y).$$

With one predictor the target slope is the population covariance over the predictor's variance, the population version of Chapter 7's covariance-to-slope identity; with several, it is the population version of the normal equations. The target is well defined, but it is an average: the projection slope of income on prestige in the additive model averages the curve's steeper and flatter stretches, weighted by where the occupations' prestige values fall. Its value therefore depends on the distribution of the predictors, and a sample with a different prestige mix would have a different target even with the same curve. It is a declared alternative target, not a rescue for a misspecified mean. HC3 standard errors remain appropriate for it under independent sampling, because the curvature the linear fit misses behaves like one more source of unequal spread (Fox, 2016).

Appendix B: Bootstrap Schemes and Intervals

Section 6 used case resampling and said why. Table 8 sets it beside the other two schemes on the same model, the same three targets, the same seed, and $B = 2000$.

Table 8

Three bootstrap schemes for the full occupation model, with each scheme’s bootstrap SE for three targets (seed 90909, $B = 2000$).

Scheme	What is resampled	Assumption carried	Education	prof – wc	prof
			slope		coefficient
Case	Whole records	Records independent and exchangeable	0.3699	1.1187	1.9926
Residual	Rescaled residuals, moved across records; design fixed	One error distribution for every record	0.2820	1.1081	1.4861
Wild	Each record’s own rescaled residual, sign flipped at random; design fixed	Each record keeps its own error scale	0.3673	1.1151	1.9767

Both design-fixed schemes rebuild outcomes from the fitted values. Residual resampling draws leverage-rescaled residuals $e_i/\sqrt{1-h_{ii}}$ with replacement and adds them to the fitted values, which treats the rescaled residuals as exchangeable across rows. The wild bootstrap keeps each row’s own rescaled residual and multiplies it by ± 1 with equal probability, so each row retains its own error scale.

Read the table by column. For the education slope the residual scheme is the outlier, 0.2820, close to the classical 0.2752, which is what one should expect, since both assume one common error distribution. The wild bootstrap, which does not, lands at 0.3673, near the case result and near HC3’s 0.3823. For the professional coefficient the pattern repeats more strongly: 1.9926, 1.4861, and 1.9767. When the schemes disagree, the disagreement is about the error model, not about the data, and it is a reason to say which scheme was used.

Four interval definitions are worth distinguishing in words.

- The **percentile interval** takes the 2.5th and 97.5th percentiles of the resampling distribution of the estimate.
- The **normal-approximation interval** takes the original estimate plus and minus 1.96 bootstrap standard errors. It uses the resampling distribution only for its spread, so it is insensitive to skewness.

- The **bootstrap-*t* interval** resamples a studentized quantity, the resampled estimate minus the original estimate, divided by the resample’s own standard error, and inverts the empirical quantiles of that ratio. It needs a standard error inside every resample, and it is more accurate when the studentized quantity is closer to pivotal than the estimate itself.
- The **BCa interval** adjusts the percentile endpoints for median bias and for an acceleration constant that captures how the standard error changes with the parameter. It is the usual recommendation when the resampling distribution is visibly skewed (Davison & Hinkley, 1997; Efron & Tibshirani, 1993).

For the three targets of Section 6 the skewness values -0.3891 , 0.2635 , and 0.3785 are mild, so percentile and normal-approximation endpoints agree closely and the higher-order corrections would change little. That agreement is a property of this example, not a general licence.

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